

3D Printing Gear Mechanism Using Fusion 360

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1. Introduction

This project explains the design and development of a 3D printed gear mechanism using Fusion 360. The model demonstrates how rotational motion is transferred between gears. The design includes two gears mounted on a custom base plate for stability and smooth operation.

2. Project Objectives

The main goal was to design a simple and functional mechanical system that could be printed easily using a 3D printer. The project also aimed to improve understanding of CAD modeling and gear motion.

3. Software Used

Fusion 360 was used for creating sketches, generating 3D models, and assembling the parts. The software provided accurate measurements and helped improve the overall design quality.

4. Gear Design

The gears were created with proper tooth spacing and alignment to ensure smooth meshing. The larger gear transfers motion to the smaller gear, demonstrating speed and torque variation.

5. Base Structure

A custom support base was designed to hold the gears securely in place. The base improves balance and keeps the shafts aligned during rotation.

6. 3D Printing Process

The final design was exported as an STL file and prepared using slicing software. Printing settings such as layer height, infill, and print speed were optimized for better quality.

7. Assembly and Testing

After printing, the gears and support pins were assembled carefully. The mechanism was tested to check smooth rotation and proper engagement between the gear teeth.

8. Advantages

The model is lightweight, low-cost, and useful for educational purposes. It demonstrates basic mechanical engineering concepts in a practical and visual way.

9. Conclusion

The project successfully demonstrated the use of Fusion 360 for creating a functional gear mechanism. The final model showed accurate motion transfer and provided valuable experience in 3D design and additive manufacturing.

Model Images



